

Requirements – Level B

Acme Starters, Inc. has requested several additional requirements that are described in this document. The new version of the project must be code named “**Acme-Starters-B**”.

Requirements

Information requirements

Implement the data model implemented in “model-B.uxf”.

Functional requirements

A **manager** must be able to:

- Create, update, and delete their own projects (only if not published).
- Select, list, and display the inventors, spokespeople, and fundraisers that are members of each of their projects.
- Publish one of their own projects, provided it has at least one invention linked to it. This implies the publishing of its components. However, prior to publishing, all component-level validations must be satisfied; if any validation fails for any component, the publication of the entire project will be aborted.
- Display any of their own projects and their components regardless of publication status.
- Display a dashboard with the following indicators: the total number of projects that they manage; the deviation of that total number with respect to the average number of projects managed by the other managers; the minimum, the maximum, the average, and the deviation of the effort required by his or her projects.

A **project member** must be able to:

- Display any of the projects that they are a member of and their components regardless of publication status.
- Add their own inventions, campaigns, and strategies to any project they are a member of, even if such components are not published yet.

Any principal must be able to:

- List and display the **projects** in the system as long as they have been published. The principal must be able to navigate to their **components** and display them.

A **sponsor** must be able to:

- Attach any of their published sponsorships to any published project.

A **auditor** must be able to:

- Attach any of their published audit reports to any published project.

An **administrator** must be able to:

- Create, update, delete, and display ad banners.

Non-functional requirements

- The user interface must be available in English and Spanish. Boolean, moment-related, money-related, and numeric data must be properly internationalised.
- Ad banners must be displayed as little intrusively as possible.

Testing requirements

- Produce sample data to cover the test cases derived from the functional requirements above. There must be three user accounts with role **manager** and credentials “**manager_i**/ **manager_i**”, where “*i*” ranges in interval one up to three. The last user account must not have any associated data, but the profile data required to create the user account itself and the corresponding role.
- Perform informal testing by navigating through the system and verifying that the functional requirements are met as expected.
- Perform formal testing by means of automated test cases that cover the functional requirements above, including:
 - Positive test cases, in which valid data is provided and the expected result is correct.
 - Negative test cases, in which invalid data is provided and the system rejects it correctly.
 - Hacking test cases, in which unauthorized access or manipulation of the system is attempted.

Managerial requirements

Intentionally blank.

Audit requirements

The customer's auditor requires the following documents: a planning report, an analysis report (including a UML data model), a Lint report, and a testing report.